

Syed Irfan M

irfuijukas@gmail.com — +91-8940742364 — GitHub — LinkedIn — Portfolio

Education

SRM Institute of Science and Technology, Trichy 2027
B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning)
5th Semester, Current CGPA: 8.00

Experience

APPROTECH 2025
Machine Learning Intern
Company: APPROTECH

- Gained practical experience in Machine Learning workflows and application development.

Projects

Sign Language to Text Conversion 2025

- Built a computer vision system to convert hand gestures into text using MediaPipe and machine learning models.
- Extracted 21 hand landmark features and applied Random Forest classifier for gesture recognition.
- Enabled real-time predictions with low latency using standard webcam input.

Network Intrusion Detection System 2025

- Built a deep learning model using CNN and LSTM to detect cyber attacks (DoS, Probe, R2L) on the NSL-KDD dataset.
- Applied SMOTE and ADASYN techniques to handle class imbalance and improve minority class detection.
- Learned spatial and temporal traffic patterns to enhance intrusion detection performance.

Nagarvaani – Tamil AI Communication Ecosystem 2025

- Built an AI-powered system integrating chatbot, voicebot, and civic complaint management for regional governance.
- Implemented speech-to-text using Whisper ASR and NLP-based classification for automated complaint categorization.
- Designed an interactive Gradio UI for real-time user interaction and complaint handling.

AgriConnect – Agricultural Supply Chain Platform 2025

- Developed a real-time platform connecting farmers, buyers, and experts using intelligent decision-based matching.
- Implemented multi-factor filtering (crop, price, urgency) to optimize supply chain efficiency.
- Integrated Firebase for real-time data synchronization across multiple users.

Publications

AICTE-VAANI, Modern Tamil Computing Summit 2025
Nagarvaani: Tamil AI Communication Ecosystem

- Developed AI framework featuring Tamil Chatbot, Voicebot, and Civic Complaint Management System.
- Implemented Whisper ASR and NLP-based automated complaint classification.
- Designed responsive Gradio UI for seamless governance support.

ICTMIM Conference 2026
Sign Language to Text Conversion

- Engineered real-time system converting sign alphabets and digits into text.

- Utilized MediaPipe hand landmarks and Random Forest Classifier.
- Designed for low-latency performance using standard webcams.

Hackathons

YUVA Hackathon (AI & Language Tech sector) – SRMIST 2025

- Developed a Smart City Infrastructure Optimizer using AI and geospatial analysis (GeoPandas) with an interactive dashboard to visualize population density and traffic congestion.
- Implemented scenario-based planning and integrated LLMs (GPT-4/Gemini) for automated recommendations; built Flask APIs and generated data-driven reports with cost-benefit insights.

BIT V-PRAYUKTI'25 – National Level AIML Hackathon 2025

- Developed a mood-aware conversational AI system using NLP and sentiment analysis to detect user emotions from text/voice inputs.
- Implemented dynamic question generation using LLMs (GPT-based) to adapt conversations and improve user engagement based on emotional context.
- Designed the system to support mental wellness and personalized interaction by guiding users toward constructive and positive dialogue.

Certifications

- RHCSA (Red Hat Certified System Administrator) – Red Hat
- Introduction to Large Language Models – Google Cloud
- Intro to Generative AI – Google Cloud
- AI Infrastructure and Operations Fundamentals – NVIDIA
- Database Structures and Management with MySQL – Meta

Technical Skills

- **Languages:** Python, Java, C
- **Machine Learning:** TensorFlow, Keras, Scikit-learn, CNN, LSTM, NumPy, Pandas, NLP
- **Web:** HTML, CSS, React.js, Node.js
- **Database:** MySQL, SQLite
- **CS Fundamentals:** Data Structures & Algorithms, OOP, Operating Systems
- **Systems & Certifications:** Linux, Red Hat Enterprise Linux, RHCSA (Red Hat Certified System Administrator)
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook, Gradio, MediaPipe, Whisper ASR, Firebase, Figma